

HECTOR GENARO PACHECO

Software Engineer | Full-Stack • Cloud Infrastructure • AI • IoT • Embedded Systems

Southern California • hector.genaro.pacheco@gmail.com • Portfolio • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

Software Engineer with a physical-world systems background spanning full-stack development, AWS cloud infrastructure, deterministic underwriting engines, retrieval-augmented AI systems, embedded Linux (Raspberry Pi), and industrial instrumentation & controls. Builds auditable, production-grade software across containerized cloud SaaS, native Android applications, and hardware-integrated event kiosk platforms.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++, Kotlin

Frontend & Mobile: React, Next.js, Android (Kotlin), Jetpack Compose, HTML5, CSS3

Backend & APIs: Node.js, Express, REST APIs, GraphQL, FastAPI, WebSockets, Zod

Systems & Hardware: Raspberry Pi 5, gphoto2, DSLR Automation, Touchscreen, systemd, GPIO, 4–20 mA, PLC

Cloud & DevOps: AWS (ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC), Docker, Linux, Git, Vercel

Data & Databases: PostgreSQL, Prisma ORM, MongoDB, Supabase, pgvector

AI & Data Platforms: OpenAI API, RAG, Embeddings, pgvector, Palantir Foundry, Palantir AIP

Quality & Testing: Vitest, Unit Testing, Integration Testing, Fault Isolation, Calibration Auditing

FEATURED SOFTWARE & SYSTEMS ENGINEERING PROJECTS

CalTrack — Industrial Calibration SaaS & Mobile Platform

[LIVE APP](#)

React • TypeScript • Node • Express • PostgreSQL • AWS ECS • Kotlin / Android

[GITHUB](#)

- Architected and deployed a full-stack industrial calibration SaaS platform managing instrument tags, calibration logs, RBAC, and auditable history snapshots.
- Containerized Node/Express backend with Docker on AWS ECS Fargate, ECR, RDS PostgreSQL, Secrets Manager, KMS, and Vercel Edge.
- Engineered a native Android field companion application in Kotlin with Jetpack Compose, ViewModels, CameraX, and ML Kit barcode scanning.

EstateOS — Real Estate Investment Intelligence Platform

[CASE STUDY](#)

Next.js 16 • React • TypeScript • PostgreSQL • Prisma ORM • Vitest • TailwindCSS

- Built a full-stack real estate analysis platform featuring deterministic financial underwriting, debt capital stacks, LTV/CLTV, and equity calculations.
- Engineered multi-strategy screening (Wholesale, Fix & Flip, BRRRR, Buy & Hold) with pure TypeScript calculators and strict input validation.
- Architected identity-bound evidence snapshots (PropertyEvidenceSnapshot) and source authority classifiers, verified by an 804-test suite in Vitest.

Interactive Photo Booth System — Client Hardware/Software

Raspberry Pi 5 • Linux • Python • photobooth-app • gphoto2 • Canon DSLR • systemd

Integration

[CASE STUDY](#)

- Built an event photo booth system for a nonprofit client, integrating Raspberry Pi 5, active cooling, an Acer 15.6" touchscreen, and tethered Canon DSLR.
- Automated camera capture via gphoto2 over USB PTP and configured systemd user services with kiosk startup scripts for reliable event operation.

FieldTrack AI — Edge Computing & Sensor Telemetry Platform

[GITHUB](#)

Raspberry Pi 5 • Python • FastAPI • Node.js • React • WebSockets • GPS/NMEA

- Built an edge telemetry platform integrating Python sensor daemons, GPS/NMEA stream parsing, motion detection, and real-time React dashboard updates over WebSockets.

Additional Repositories & Deployments: Procurement Tracking — Full-Stack GraphQL Enterprise App (React/Apollo/Prisma/PostgreSQL)

[LIVE APP](#)

[GITHUB](#)

• **OC Water Features** — Production Web Platform (Next.js/Supabase/Vercel)

[LIVE SITE](#)

• **Pi Arcade OS** — Embedded System (Python/GPIO)

[SITE](#)

[GITHUB](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Instrumentation & Controls Technician — Process Control & Industrial Systems

Field Engineering Experience

- Calibrated, commissioned, and troubleshot 4–20 mA process instrumentation (pressure, temperature, flow, level), PLC I/O wiring harnesses, relays, and field sensors across 480VAC, 120VAC, and 24VDC power systems.
- Applied systematic fault isolation, signal integrity validation, and root-cause diagnosis in operating industrial plant facilities while maintaining auditable calibration logs.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026, 16.00 Continued Education Credits)